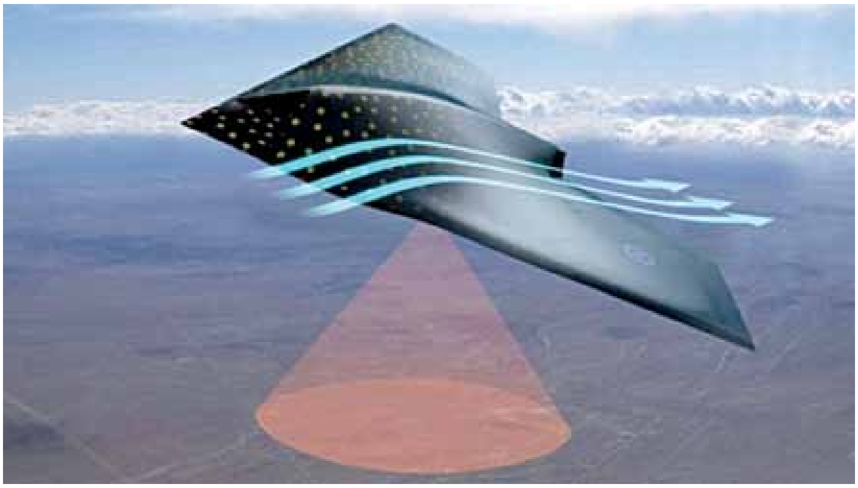
lation of a large number of sensors, without any restrictions on their location, hence providing additional or more precise data. The new information, provided without any additional wires, allows for an improved control of the aircraft and presents major advantages compared to traditional solutions, notably in terms of fuel consumption and maintenance costs.

The sensors will have a life of 35 years. With this unlimited autonomy, a wide array of sensors can now be placed very close to the structures to be monitored, at places that would be very difficult to access for maintenance, such as in the fuselage, the wings, and the stabiliser. Each sensing node allows the measurements to be timed with an accuracy of less than a millisecond, for sampling frequencies ranging from 120 to 500Hz. The information is communicated to the plane computers by radio. They do not interfere with the operation of the pre-existing devices, as the system successfully passed the qualification tests related to radio interferences.

Courtesy: BAE Systems



Courtesy: BAE Systems

Innovative health management and navigation solutions for helicopters

The project initiated by Airbus as part of the 'Clean Sky' programme resulted in the development of a high-performance and ultra-low-power wireless sensor network platform able to monitor aircraft critical structures. Each sensor is equipped with a radio transceiver and an independent power supply. Energy consumption, both during flight and while the aircraft is on the ground, has been reduced to a minimum by a combination of energy-efficient electronic components and by the optimisation of the communication protocol.

By combining the outputs of thousands of wireless sensors with big data analysis, the technology will benefit more robust defense platforms that are capable of complex missions whilst reducing the need for routine maintenance checks.

Using the smart skin technology would even reduce the need for excessive ground check-ups and allow for quick part replacement, if needed. Overall, the system could increase aircraft maintenance efficiency and improve safety. The tiny sensors that would be used in the technology could be as small as grains of rice or even as small as dust particles at less than 1mm squared. The sensors can be retrofitted to existing aircraft and spraying them like paint.

Innovative health management and navigation solutions for helicopters

The distributed rotorcraft wireless health monitoring solutions provide a significant amount of information regarding health and usage of helicopters.

These combine strain gauge signal conditioning, temperature sensors, and tri-axial accelerometer in an environmentally hardened form factor. Also, high-performance miniature inertial measurement unit and vertical gyro and ruggedised tactical grade GPS-aided inertial navigation system for precise landing.

These synchronised wireless sen-